Chiara Tramarin

MSc in Physics of Data
Vicenza, Italy

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ChiaTrama
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Education

•Master's Degree in Physics of Data Mar. 2024 – Oct. 2026
University of Padua, Padua, Italy Grade: 110/110

•Bachelor's Degree in Physics Oct. 2018 – Apr. 2024
University of Padua, Padua, Italy Grade: 92/110
Erasmus exchange: Universidad Autónoma de Madrid, Spain · Sept. 2022 – Mar. 2023.

•Scientific High School Diploma 2018
Liceo P. Lioy, Vicenza, Italy

Research Experience

•Research Intern – IFIMAC, Madrid Mar. – Sept. 2026

Conducted computational research on DNA mechanics and nucleosome stability using coarse-grained molecular dynamics simulations. Performed free-energy calculations through thermodynamic integration and analyzed simulation data using Python to investigate sequence-dependent physical properties. Ran simulations on an HPC GPU cluster using Linux, Bash and SLURM.

Master's thesis: *Investigating sequence-dependent DNA mechanics and nucleosome stability through coarse-grained simulations.*

•Bachelor's Thesis Research – University of Padua Nov. 2023 – Apr. 2024

Performed experimental measurements on a lithium-niobate ($LiNbO_3$) opto-microfluidic Lab-on-a-Chip for detecting transparent microplastics in water droplets, combining sample preparation, optical and electronic instrumentation (lasers, oscilloscopes and microfluidic control), and Python-based data processing and visualization.

Bachelor's thesis: *Optical sensing study of microplastics in droplets via refractive index contrast.*

Co-author of a related peer-reviewed publication: *Optofluidic light-droplet interaction for rapidly assessing the presence of plastic microspheres within aqueous suspensions. Optics Express*, 34(5), 8175–8194 (2026).

Selected Academic Projects

•N-Body Simulation on CPUs and GPUs
Developed a C++ gravitational N-body simulator with serial, OpenMP and CUDA implementations; validated numerical accuracy and energy conservation, and benchmarked CPU/GPU performance.

•Bayesian Optimization
Applied Gaussian-process Bayesian optimization to optimize synthetic black-box functions and CNN hyperparameters, comparing acquisition strategies and evaluating performance against random search.

•Simulation-Based Inference for a 2D Random Walk
Applied simulation-based inference with neural posterior estimation to recover parameters of a stochastic 2D random walk, validated on an analytically solvable quartic model, and extended to a red-blood-cell model not accessible analytically. Optimized simulation performance with Numba to achieve C-like speed.

•Car Image Classification with CNNs
Built and evaluated PyTorch CNNs for fine-grained vehicle classification and Siamese verification on CompCars, applying transfer learning, fine-tuning and strategies for severe class imbalance.

•Distributed K-Means with Dask
Implemented and benchmarked a distributed K-Means‖ algorithm with Dask, extending the dask-ml implementation with weighted centroid updates. Configured VM clusters via SSH, optimized memory and thread allocation, and analyzed scalability on large datasets (KDD Cup 1999).

•Complex Networks: Social Connectedness and Self-Organized Criticality
Analyzed weighted, geolocated Facebook social-connectivity data and investigated how network structure influences collective dynamics; separately simulated self-organized criticality with the Bak–Tang–Wiesenfeld model across different network topologies.

Technical Skills & Languages

Programming Languages Python, C++, R, MATLAB, SQL
High-Performance & Parallel Computing CUDA, OpenMP, Dask, Numba, Slurm
Software & Tools Git, Docker, Linux, Jupyter, VS Code, LaTeX, Microsoft Office (Excel, Word, PowerPoint)
Languages Italian (native), English (advanced), Spanish (advanced)